Maya Arnott

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Data scientist with experience in high-dimensional data, statistical modeling, and pipeline development, focused on generating insights from complex real-world health and life sciences data.

EDUCATION

Master of Science in Environmental Health Data Science *Columbia University*, August 2025 - August 2026
Relevant Coursework: Biostatistical Methods (regression modeling, hypothesis testing, ANOVA), Data Science (data visualization & data wrangling), Public Health GIS (spatial analysis & mapping), Machine Learning for Epidemiology (supervised/unsupervised learning)
Clubs: Columbia Mailman Health Tech Group

Master's Thesis with Dr. Haotian Wu, Columbia University (In Progress):

- Analyzing 500,000 chemical features and prostate cancer aggressiveness using high-dimensional clinical and exposomics datasets
- Applying PCA, multivariable regression, and interaction modeling, along with data preprocessing and visualization in Python and R

Bachelor of Arts in Environmental Science *Washington University in St. Louis,* August 2019 - May 2023
Major Environmental Biology **Minor** Urban Studies
Relevant courses: Biogeochemistry, Applications in GIS, Environmental Engineering, Computing for Engineers (Python/MATLAB & Computer Systems), Statistics, Microeconomics (Consumer Theory, Market Structures & Failures)

SKILLS & CERTIFICATIONS

- **Programming Languages:** R, Python, SQL
- **Libraries, Frameworks, Techniques:** tidyverse, Shiny, pandas, NumPy, raster, leaflet, matplotlib
- **Databases & Data Handling:** PostgreSQL, MySQL, Excel, structured datasets, APIs
- **Tools:** Git, GitHub, VS Code, RStudio, Jupyter Notebook
- **Laboratory Skills:** Flow cytometry, ELISA, BAMA, phagocytosis assays, cell culture, sterile technique, buffer formulations, SOP compliance

EXPERIENCE

Life Science & AI Research Expert, *Mercor Corporation* | Remote Contractor | May 2025 - February 2026

- Design domain-specific evaluation datasets and prompts to train and fine-tune large language models in biology and life sciences, reducing hallucination rate by 26.7%
- Assess model output quality, factual accuracy, and reasoning consistency across complex scientific tasks, collaborating with engineers and researchers to improve model performance

Research Technician II, *Duke University* | Durham, NC | July 2023- June 2025

- Automated data reporting pipelines in R and SAS to streamline experimental workflows, reducing turnaround time by 30%
- Synthesized complex quantitative results to support data-driven recommendations for research strategy, communicating findings to both technical and non-technical stakeholders
- Designed and executed rigorous experimental validation protocols to evaluate candidate performance across multiple testing conditions
- Trained junior researchers on technical procedures and analytical interpretation of assay readouts

Air Quality Analyst, *Washington University in St. Louis*| St. Louis, MO | June 2022- May 2023

- Developed custom Python pipelines to process and clean global air quality datasets exceeding millions of records from the SPARTAN particulate monitoring network, improving analytical workflow efficiency by 40%
- Applied image analysis and regression modeling to quantify black carbon concentrations, supporting peer-reviewed environmental impact assessments
- Communicated complex analytical findings to academic and engineering stakeholders, contributing to larger research publications

Research Intern, *Donald Danforth Plant Science Center*| St. Louis, MO | March 2021- June 2022

- Analyzed large datasets on plant material/phenotypes to develop sustainable crops using R
- Focused on determining the extent to which plant traits determine community and soil ecosphere properties

ACTIVITIES

Varsity Basketball Team Co-Captain, *Washington University in St. Louis* | St. Louis, MO | August 2019 - May 2023

- All American team in 2023 and received Junior Athlete of the Year Award for my skill and excellence in playing varsity basketball